Tempo

Quick reference · control software for SRS DG645 delay generators

1 Install and start

- **Windows:** double-click `Tempo.exe`. If Windows says it protected your PC, choose **More info ▸ Run anyway**.
- **USB stick:** `Tempo-Portable.exe` keeps everything in a `Tempo-data` folder beside it.
- **Linux:** install the `.deb` (Arch: the `PKGBUILD`), then start **Tempo** from the menu or run `tempo-dg645`.

2 Add your instrument

- **Instrument ▸ Add...** `Ctrl`+`N`: a name you will recognise (e.g. *Laser lab DG645*) and its IP address, from the DG645's front-panel network settings.
- Or let Tempo look: **Instrument ▸ Find on network...** `Ctrl`+`F`.
- **Test connection**, then **Save**. No hardware? **Instrument ▸ Add demo instrument**.

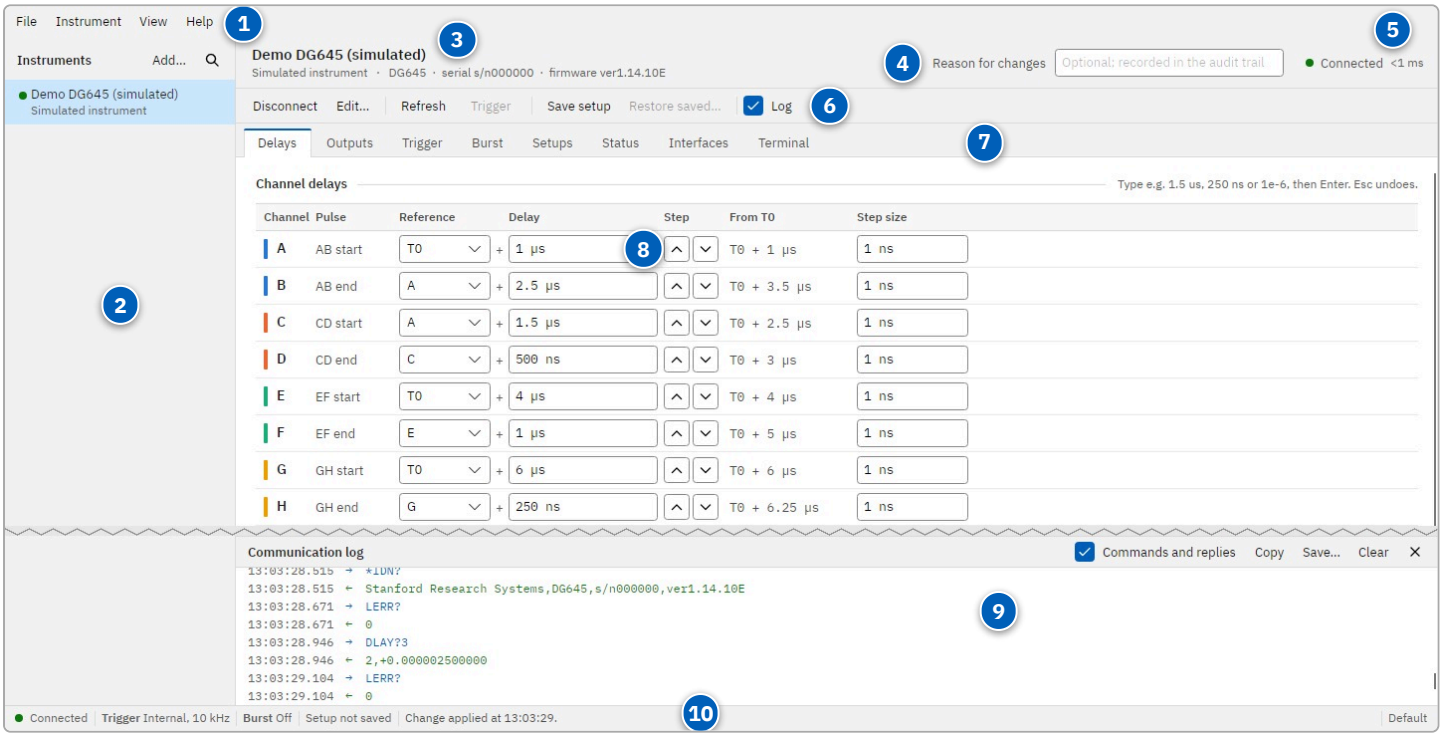
3 Connect

- Double-click the instrument in the list, or select it and press `Ctrl`+`K`. Tempo reads every setting into the pages.
- The dot turns ● green and the header says *Connected*, with the reply time. Tempo then checks the instrument every second.
- The DG645's front panel is locked while connected. **Disconnect** `Ctrl`+`Shift`+`K` hands it back.

4 Change a value

- Click a field, type the value with its unit (`1.5 us`, `250 ns`, `10 kHz`), then press `Enter` or click anywhere else.
- **Amber**: typed, not sent yet. **Red**: the instrument refused it, and the reason is underneath. `Esc` undoes.
- Every value is read back, so the screen shows what the instrument really holds.

The main window



1. **Menus.** *File*: configurations, saving setups. *Instrument*: add, connect, trigger. *View*: log, audit trail, alert sound, theme.
2. **Instruments.** Double-click to connect; right-click for more. The dot shows the connection.
3. **Header:** name, address, model, serial number and firmware.
4. **Reason for changes** (optional), recorded with every change in the audit trail.
5. **Connection and heartbeat:** *Connected* and how fast the instrument answered.
6. **Command bar:** Connect, Edit, Refresh, Trigger (or Arm), Save setup, Restore saved, Log.
7. **Pages:** Delays (with a timing diagram), Outputs, Trigger, Burst, Setups, Status, Interfaces, Terminal.
8. **Value fields.** Type and press Enter; the up and down buttons step by the step size.
9. **Communication log:** every command sent and reply received.
10. **Status bar:** state, trigger, burst, *Setup not saved*, messages, configuration.

If anything goes wrong, a coloured **alert bar** appears across the top of the window (see side 2).

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Save a setup and put it back

- **Save setup** stores the instrument's complete setup (delays, output levels, trigger and burst) on this computer. **File ▸ Save setups** `Ctrl`+`S` saves every connected instrument at once.
- **Restore saved...** `Ctrl`+`R` puts the saved setup back on the instrument.
- *Setup not saved* in the status bar, and an asterisk in the title, mean something has changed since the last save.
- Setups are kept in a **configuration** file (`.tempo`). Keep one per experiment or lab with **File ▸ New / Open / Save configuration as**. The last one opens automatically next time.

When an instrument connects

Tempo compares the instrument with its saved setup. If they differ, it lists what is different and asks you to choose:

Load saved setup puts your saved settings on the instrument.
Keep instrument settings leaves it as it is.

Tick *Remember my choice* to do the same automatically every time. You can change this later with **Edit...**.

Triggering

Choose the trigger **Source** on the **Trigger** page.

- **Single shot: Trigger** `Ctrl`+`T` fires one delay cycle.
- **Single shot, external rising/falling:** press **Arm**. The button turns into **● Ready**, on every page and in the status bar. The next edge at the trigger input fires the shot, and the button goes back to **Arm**.

How the delays work

Each channel **A–H** fires a set delay after its **reference**: T0 (the start of every cycle) or another channel. Each output pulse runs from one channel to the next: **AB** starts at A and ends at B, and likewise **CD**, **EF** and **GH**.

Example: `A = T0 + 1 us`, `B = A + 2.5 us` gives an AB pulse **2.5 µs wide**, starting 1 µs after T0. Because B follows A, moving A moves the whole pulse.

Typing values

Time	`ps` `ns` `us` (or `µs`) `ms` `s`, e.g. `250 ns`, `1.5us`, `1e-6`
Frequency	`mHz` `Hz` `kHz` `MHz`, e.g. `10 kHz`, `2.5k`
Voltage	`mV` `V`, e.g. `2.5 V`, `800 mV`, `-0.8`

A number on its own is in seconds, hertz or volts. Always use a full stop for the decimal point.

Audit trail (ISO 9001 records)

- Every change Tempo makes is recorded: who made it, when, on which instrument (with its serial number), each setting's **old → new** value, and your reason.
- Type a **Reason for changes** in the instrument header first, for example a job or change number.
- **View ▸ Audit trail...** shows the records. Filter and search them, use **Check integrity** to prove none have been altered, and use **Change folder...** to keep them on a backed-up network drive.

Heartbeat and alerts

While connected, Tempo checks the instrument **every second**. The dot by its name shows the result:

● Connected	Answering normally; the header shows the reply time (e.g. *4 ms*).
● Not responding	A check went unanswered. Usually it recovers by itself.
● Connection lost	No answer for about 10 s, or the link was closed.
○ Not connected	Double-click to connect.

Problems appear in an **alert bar** at the top of the window, with a sound (**View ▸ Alert sound**):

> ! **Not responding** · clears itself when the instrument answers again
> Check that it is switched on and its network cable is plugged in.

> × **Connection lost** · **Reconnect**
> Fix the power or cable, then press **Reconnect** in the alert.

> × **PLL unlocked · Rubidium unlocked · Timebase changed**
> The instrument's timing reference has a fault. Check the timebase and any external 10 MHz reference. Timing may have been wrong while this lasted.

Show jumps to the instrument, and **×** dismisses the alert. Alerts are also kept in the audit trail.

Keyboard shortcuts

`F1`	User guide, at the page you are on
`Ctrl`+`N` / `Ctrl`+`F`	Add an instrument / find on network
`Ctrl`+`K` / `Ctrl`+`Shift`+`K`	Connect / disconnect
`F2`	Edit the selected instrument
`Ctrl`+`S` / `Ctrl`+`R`	Save setups / restore the saved setup
`Ctrl`+`O`	Open a configuration
`F5`	Read every setting again
`Ctrl`+`T`	Trigger, or arm
`Ctrl`+`L`	Show or hide the communication log
`Enter` / `Esc`	In a field: send the value / undo

If something goes wrong

Will not connect	Check the IP address against the front panel and that the computer is on the same network. Close any other program using the instrument, and try **Find on network**.
Value refused	Read the message under the red field (e.g. *Illegal delay*). Press `Esc` to go back.
Front panel locked	**Disconnect**, or **Go to local** on the Interfaces page.
Pages may be out of date	Shown after typing commands in the Terminal. Click it or press `F5`.
Audit trail not written	The audit folder cannot be reached. Records wait and are written once it is back.